Gaurav Mehta

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Education

North Carolina State University, Raleigh, NC **08/2024 – 05/2026**
M.S. Data Science

K.J. Somaiya College of Engineering, Mumbai, India **09/2020 – 05/2024**
B.S. Computer Science

Work Experience

Data Science Intern **07/2026 – 09/2026**
ApexAnalytix, Greensboro, North Carolina

- Authored **440 red-team test cases** aligned to the OWASP Top 10 for LLMs and NIST AI 100 to test chatbot and AI agent security
- Reduced weeks of manual pre-release testing for AI agents and RAG systems by building an automated testing framework in Python and PostgreSQL, deployed with Docker, Helm Charts and Kubernetes, cutting testing time from weeks to 1-2 hours, a 95%+ reduction.
- Standardized evaluation of in-house LLMs by building a Python benchmarking framework that generated model cards for clients and drove model updates, improving AI product performance by 4%.
- Diagnosed workflow bottlenecks in client-deployed AI agents by building SQL and Python monitoring and analysis pipelines, enabling targeted fixes that improved overall product performance by 15%.

Research Assistant **05/2025 – 07/2025**
North Carolina State University, Raleigh, North Carolina

- Assessed LLM-driven document extraction accuracy across **10,000 invoice images spanning 50 templates**, identifying format variability as the key bottleneck limiting generative AI-based automation.
- Improved extraction accuracy by **23% over baseline** by benchmarking traditional OCR against baseline and fine-tuned DONUT transformer models, enabling more reliable automated document-processing pipelines.

Data Analyst Intern **01/2024 – 03/2024**
LTIMindtree, Mumbai, India

- Tackled rising customer attrition for a banking client by engineering a supervised machine learning pipeline on 1M+ customer records, achieving **84% prediction accuracy** in flagging at-risk accounts.
- Optimized model performance by comparing Random Forest and XGBoost ensemble algorithms against Logistic Regression, **lifting F1-score from 0.71 to 0.83** via hyperparameter tuning and feature selection.
- Automated feature engineering and preprocessing at scale using PySpark and AWS Glue, **cutting pipeline runtime by 35%** and enabling continuous, automated model retraining.
- Built a QuickSight reporting pipeline surfacing churn risk scores for **50,000+ accounts** to 4-5 business stakeholders, directly informing data-driven retention strategy.

Projects

Sales Analytics Data Pipeline

- Engineered a scalable, end-to-end ELT pipeline using Python, SQL, and dbt with dimensional data modeling, transforming raw sales and marketing data into **10+ production-ready fact and dimension tables** for analytics at scale.
- Delivered **3 interactive Tableau dashboards** powering automated data validation and metric governance, enabling self-service business intelligence across the analytics workflow.

Federated Learning Data Orchestration

- Orchestrated a distributed, fault-tolerant data pipeline using Python, Bash, and GitHub Actions to simulate federated learning workflows, automating ingestion and preprocessing of **200M+ medical records** across privacy-sensitive environments.
- Implemented automated schema validation and data quality checks within a CI/CD-driven MLOps pipeline, **reducing errors by 13%** and enabling reproducible, version-controlled model training workflows.

F1 Telemetry Dashboard (Real-Time Analytics)

- Designed a real-time ETL pipeline using Pandas and NumPy to process **100K+ telemetry records per race**, powering a low-latency Streamlit/Plotly dashboard for self-service, lap-level performance analysis.

Skills

- **Languages** : Python, SQL, R, Java, JavaScript
- **Machine Learning & AI**: PyTorch, Scikit-learn, Hugging Face, Transformers, XGBoost, Random Forest, NLP, LLM Evaluation
- **Data Engineering**: PySpark, dbt, AWS Glue, Pandas, NumPy, Dimensional Data Modeling
- **MLOps & DevOps**: Docker, Kubernetes, Helm Charts, FluxCD, Git, GitHub Actions
- **Databases & Cloud**: PostgreSQL, MySQL, MongoDB, AWS
- **Analytics & Visualization**: Tableau, Power BI, QuickSight, Streamlit, Plotly, Excel
- **Certifications**: Natural Language Processing with Sequence Models; Natural Language Processing with Probabilistic Models (Coursera)